



Research Paper

Heat shock protein 70 regulates m6A modification in response to heat shock in esophageal squamous cell carcinoma

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ABSTRACT

Heat shock has been known to induce hyperplasia in esophageal epithelial cells. It is widely considered as a crucial risk factor in the initiation and development of esophageal squamous cell carcinoma (ESCC), yet our understanding of the underlying mechanisms remains limited. The m6A modification of mRNA plays a role in mediating several cellular processes and is critical during cell stress. Our study revealed that inhibiting of m6A 'writer' components of ESCC cells exhibit higher death rates and slower recovery after heat shock. After normalization using mRNA expression profiles, 91.08 % of significantly changed m6A modifications aligned with corresponding mRNA abundance changes, with no evidence of over-modification, while the increase in m6A modification of 8.92 % of heat-shock associated genes far exceeded the increase in mRNA (hyper - m6A modification), and A/U rich motifs were commonly observed in the 3'UTR of these gene. Inside the nucleus, the binding of HSP70s in m6A writer complex promote the hyper - m6A modification in specific mRNAs after heat shock. The stronger nuclear localization of HSP70 in ESCC tissues correlates with a poor prognosis for the patients. In conclusion, our research revealed that the nuclear HSP70 protein could bind to the METTL3/14 writer complex and regulate mRNA's m6A modification. Our results provide a new perspective for research into how HSP70 protein regulates mRNA stability and suggests a new direction for the comprehensive prevention and treatment of ESCC.

0. Introduction

Esophageal carcinoma (EC) is the eighth most prevalent and the sixth most deadly cancer globally [1]. Due to the limited clinical approaches for early diagnosis, EC tends to have a poor prognosis. The 5-year survival rate of EC ranges from 10 % to 25 % [2]. Approximately 90–95 % of the EC patients were diagnosed with ESCC in the Taihang Mountains of north-central China, a region that accounts for the highest ESCC incidence in the world [3]. However, the underline mechanisms remain less clear.

Several factors were considered significantly related to the development of ESCC, including alcohol consumption, smoking, and poor oral environment [4]. Indeed, epidemiological studies show that hot drinking (with a temperature over 65 °C) is an independent risk factor for ESCC [5,6]. Studies in some regions have shown that long-term hot drinks could increase 90 % cancer risk of esophageal cancer [7].

Hot drinking could promote the progression of esophageal cancer via various candidate pathways [7], such as induction of DNA double-strand

breaks [8], base oxidation, epigenetic changes [9] and chronic inflammation [10]. However, population-based studies indict that hot drinks and heat shock could lead to abnormal proliferation of basal layer cells but are not associated with DNA damage [11]. Studies in mice have also shown that heat shock stimulation is effective in promoting the initiation and progression of ESCC [12]. However, the underling mechanisms remain less clear.

m6A RNA modification represent the most populous type of methylation modification found in cells. m6A on mRNA is formed by "writer" complex constituted by METTL3 and METTL14 in a 1:1 ratio [13]. The mRNAs carrying m6A modifications can be recognized by "reader" molecules that facilitate either the translation or degradation of these mRNAs [14]. METTL3/14 mediated m6A modification is a crucial mechanism through which cells proliferate, differentiate, and respond to various types of stress [15]. In tumorous cells, increased levels of METTL3 and METTL14 regulate mRNA metabolic processes through writing in m6A modification, driving malignant phenotypes such as cellular survival, proliferation, self-renewal, metastasis, and drug

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resistance [16]. Various mechanisms underpin the modification of m6A by the writer complex, including the transcription factor-coupled insertion [17], direct binding to methylated histones [18], guiding by different RNA-binding proteins, and random modifications [19]. These differential modification mechanisms determine the diversity of the m6A modification spectrum under different life processes and stress conditions.

As an important factor in the onset and progression of ESCC, heat shock can also cause drastic changes in m6A modifications, thereby regulating the expression and translation of mRNA within cells, and affecting the cells' ability to adapt to heat shock. Some studies have described changes in m6A modifications within cells after heat shock, where many mRNAs still exhibit significant upregulation of m6A modification, even after excluding differences in mRNA levels (hyper-m6A modification) [20–22]. A possible explanation for this phenomenon is provided by the “reader” of m6A, YTHDF2, which is translocated from the cytoplasm to the nucleus after heat shock, and competes for the m6A modification site on mRNA with FTO [23].

However, these phenomena have not been adequately researched. For instance, while YTHDF2 preferentially binds with the methylated site at 5'UTR, increases in m6A modification locations are predominantly found at 3'UTR and CDS regions. The knockdown of YTHDF2 only affects the translation of heat shock proteins but does not affect the lifespan of the relevant mRNA [24]. Therefore, identifying molecules that affect the writing preference of the METTL3/14 complex may provide significant insights into elucidating this mechanism.

Following heat shock, changes in m6A modifications have been reported to promote the migration of sublethal cells to the lungs after hepatic radiofrequency ablation surgery [25]. This highlights the importance of elucidating the unique m6A modification mechanisms post heat shock for both the prevention of the initiation and development of cancer.

In this study, we subjected cells to heat shock treatment and identified mRNA with hyper-m6A modification after the heat shock through multi-omics approaches. Using proteomics, we investigated the differences in proteins bound to the METTL3-METTL14 complex before and after heat shock and identified the guiding role of the METTL3-METTL14 complex-bound protein HSP70 in m6A ‘writing’ preferences. This will help us understand the function of the nucleus-localized HSP70 protein in tissue cells.

1. Materials and methods

1.1. Cell culture and reagents

EC109 ESCC cell lines were purchased from National Laboratory cell resource sharing platform (Beijing). EC109 were recovered and cultured with DMEM (Gibco, Grand Island, NY) medium containing 10 % bovine serum (Gibco, Grand Island, NY) until the cells were > 80 % confluent. For heat shock, cells were soaked in a water bath at 45/55/65 °C for 30 min. The samples were subjected to thermal treatment using a circulating water bath equipped with a temperature control module (± 0.1 °C). Following incubation, the cell culture dishes were removed, equilibrated at ambient temperature for 5 mins, and subjected to surface sterilization with 70 % ethanol. Subsequently, the pre-warmed complete medium was replenished, and the dishes were placed in a 37 °C, 5 % CO₂ incubator. HSP70 inhibitor VER-155008 were purchased from MedChem Express (HY-10941, China). STM2457 were purchased from MedChem Express (HY-134836, China).

1.2. Annexin V/PI analysis

Cells were treated with heat shock (45 °C, 30 min), followed by a 24 h recovery with or without STM2457. Apoptosis was analyzed using an Annexin V-FITC/PI dual staining kit (C1062M, Beyotime) following standard protocols. Briefly, cells were harvested, washed twice with cold

PBS and resuspended in 100 μ L of Annexin V Binding Buffer. Cells were then stained with 5 μ L FITC-conjugated Annexin V and 5 μ L PI for 15 min at room temperature in the dark. Prior to flow cytometry analysis, samples were diluted with 400 μ L binding buffer and immediately analyzed using 488 nm excitation with emission filters at 530 nm (FITC) and 617 nm (PI).

1.3. Western blot

RIPA (Beyotime, Shanghai, China) lysis buffer was used to extract the total protein from esophageal carcinoma cells. Secondary antibodies raised against rabbit or mouse IgG 1:5000 were incubated at 4 °C for 12 h after washing the membranes six times with TBST for five minutes each. ChemiDOCTM XRS + imager Bio-Rad, and protein levels were quantified using the Olympus Image-Pro Plus software. Immunoblots were scanned by the equipment. Antibody used for western blot were as follows: anti-METTL3 (Abcam, ab195352), anti-HSP70 (CST, 4873), anti-beta-actin antibody (Abcam, ab8226), Goat Anti-Rabbit IgG H&L (HRP) (Abcam, ab205718), Goat Anti-Mouse IgG H&L (HRP) (Abcam, ab205719).

1.4. meRIP-seq and mRNA-seq

The execution of MeRIP-Seq was handed over to Cloudseq Biotech Inc. (Shanghai, China), adhering to the established protocol (Meyer et al., 2012) with minor adjustments. In short, the fragmented RNA was treated with an anti-m6A polyclonal antibody (Synaptic Systems, 202003) in IPP buffer and kept for 2 h at 4 °C. Following this, the RNA-antibody mix was immunoprecipitated by combining it with protein-A beads (Thermo Fisher) and further incubated at 4 °C for another 2 h. The RNA bound to the beads was then extracted using N6-methyladenosine (BERRY & ASSOCIATES, PR3732) in IPP buffer and was isolated using Trizol reagent (Thermo Fisher) according to the provider's guidelines. This resultant purified RNA was used to generate an RNA-seq library utilizing the NEBNext® Ultra™ II Directional RNA Library Prep Kit (NEB). Both the initial sample and the m6A IP samples, both with and without immunoprecipitation, were put through the process of 150 bp paired-end sequencing on an Illumina HiSeq sequencer.

The Illumina HiSeq 4000 sequencer produced paired-end reads that underwent a Q30 quality control assessment. The cutadapt software (v1.9.3) was utilized to trim off the 3' adaptor and discard low-quality reads. The resulting reads were mapped to the reference genome (UCSC HG19) using the Hisat2 software (v2.0.4). The MACS software (v1.4.2) was deployed for m6A peak identification, while DiffReps software pinpointed differentially enriched m6A peaks (Fold change cut-off: 2, P-value cut-off: 0.00001). Peak annotation was done using in-house scripts. Genes with altered methylation underwent GO and Pathway enrichment analysis. The IGV tool, widely used for this purpose, enabled visualisation of read alignments on the genome.

1.5. Real-time quantitative PCR

Total RNA was extracted using the TRIzol reagent (Takara, 9109). The High Capacity cDNA Reverse Transcription Kit (Takara, 6110 A) was used to carry out reverse transcription. We conducted Real-Time PCR analysis using BeyoFast™ SYBR Green qPCR Mix (Beyotime, D7260) on a LightCycler 480 Real-Time PCR System (Roche Applied Science). Specified below are the primers utilized for amplifying each target.

GAPDH

Fwd: 5'-GGAGCGAGATCCCTCCAAAAT-3';

Rev.: 5'-GGCTGTTGTCATACTTCTCATGG-3'.

HSPA1B

Fwd: 5'-TTTGAGGGCATCGACTTCTACA-3';

Rev.: 5'-CCAGGACCAGGTCGTGAATC-3'.

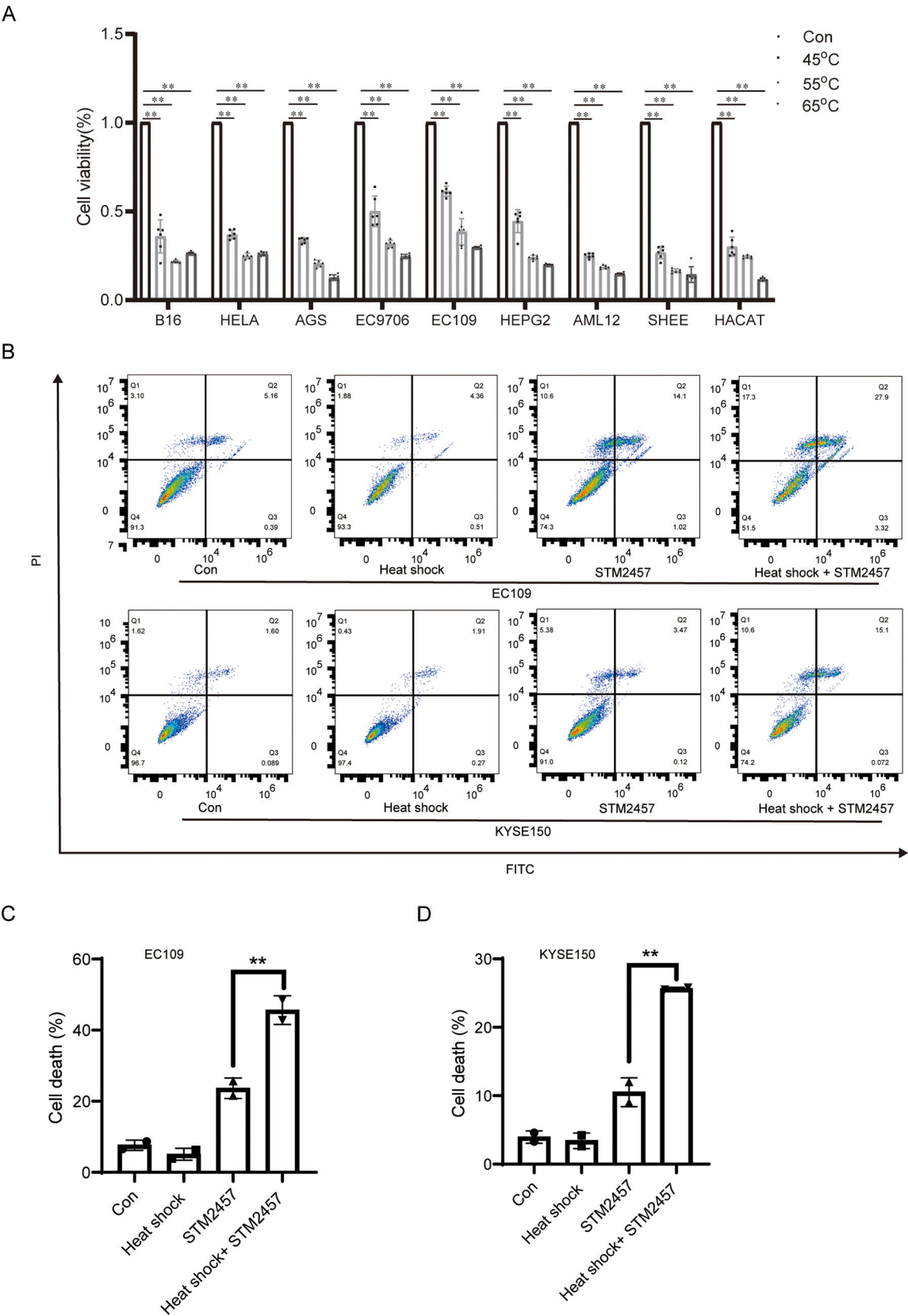


Fig. 1. Inhibiting m6A modification promote cell death during heat shock. (A) Cell lines challenge with heat shock under 45, 55, 65 °C for 30 min, after 24 h recovery, cell viability was measured with MTT analysis($n = 6$); (B) EC109 and KYSE150 cell were treated with heat shock (45 °C, 30 min) and recovery for 24 h with or without m6A inhibitor STM2457(100 μ M). Cell death was valued through Annexin V-FITC/PI staining; (C) Cell death percent of EC109 cell; (D) Cell death percent of KYSE150 cell. The data were presented as mean $\pm$ SD, one way ANOVA was used for the analysis, $n = 2$, $**p < 0.01$.

1.6. Immunofluorescence staining

We cultured ESCC cells on coverslips and fixed them in 4 % paraformaldehyde after washing three times in 0.1 M PBS for 3 min each. A bovine serum albumin solution was immersed in the sample for 60 min at room temperature and then incubated at 4 °C overnight with anti-HSP70 antibody and anti-METTL3 antibody. Coverslips were washed 3 times in PBS for 3 min each, then incubated for 60 min with a secondary antibody. Goat Anti-Rabbit IgG H&L (Alexa Fluor® 488, Abcam, ab150077, 1:1000) and Goat Anti-Mouse IgG H&L (Alexa Fluor® 594, Abcam, ab150116, 1:1000) were used as the secondary antibody for IF staining. Finally, histology mounting medium (Sigma, F6057) were used to mounting the slides. Following immunostaining, the cells were examined under a light microscope (Olympus, IX73).

1.7. LC-MS/MS analysis

Analyses were performed on a Q Exactive HF-X mass spectrometer coupled with a Thermo Fisher Scientific Easy LLC 1200. The phosphopeptides were loaded onto a self-packed column using buffer A for the phosphoproteomic study. At a flow rate of 300 nL/min, peptides were eluted over 110 min using a linear gradient of buffer B from 2 to 40 %. MS scans were acquired from m/z 350 to m/z 1800 with a resolution of 60,000 at m/z 200 and an injection time of 50 ms. Afterwards, data-dependent top 15 MS/MS scans with normalized energy 28 were applied with 15,000 resolutions at 200 m/z using higher-energy collision dissociation (HCD). Isolation window size was set to 1.6 Th, and dynamic exclusion duration was 30 s.

1.8. ESCC tissue microarray and IHC staining

ESCC tissue microarrays were purchased from Shanghai Outdo Biotech in Shanghai, China (HEsoS105Su01). For immunohistochemistry staining, antibodies against HSP70 were used. The immunohistochemistry kit was purchased from Sangon Biotech (D601037). For measuring the percentage of positive cells and the intensity of the stain, we quantitatively scored the tissue slides under a microscope. In a blinded study, two pathologists validated the scoring system independently.

1.9. Co-immunoprecipitation

Firstly, cells from different treatments were scraped off using a cell scraper and centrifuged at 4000 rpm for 2 min at 4 °C. The cell pellet was then lysed with 500 μ L of RIPA lysis buffer (Beyotime, P0013D). The lysate was thoroughly mixed with antibody-bound magnetic beads (Beyotime, P2173) and incubated overnight on a rotator at 4 °C. The following day, the magnetic beads were washed three times with RIPA lysis buffer. Next, 50 μ L of 2 $\times$ loading buffer was added, followed by a 10-min denaturation process at 95 °C to separate the proteins from the magnetic beads.

1.10. RNA stability assay

Cells were treated with 5 μ g/mL actinomycin D to inhibit transcription, with samples collected at 0, 2, 4, 6 h for RNA extraction using TRIzol reagent. Total RNA was reverse-transcribed into cDNA and quantified via SYBR Green-based qPCR, normalized to spike-in control using the $2^{-\Delta\Delta Ct}$ method. Biological triplicates and statistical thresholds ensured reproducibility.

1.11. Statistics

In this study, student's *t*-test was used for the analysis between two groups of data, while one-way ANOVA was employed for the analysis of multiple groups of data. The data were presented as mean $\pm$ standard

deviation.

1.12. Statement

In the preparation of this manuscript, we utilized Deepseek R1 for limited purposes, specifically for grammar correction and language enhancement. It is important to note that no content was directly generated by the AI. The manuscript is finally reviewed by a author to ensure scientific accuracy.

2. Result

2.1. m6A promote heat shock response of ESCC cell line

To systematically evaluate cellular thermotolerance under heat-shock, we performed comparative analyses across diverse cancer cell lines and immortalized cell models (Fig. 1A). Notably, ESCC cell lines displayed significantly enhanced survival rates post heat exposure compared to other tested cell types. m6A modification is the most abundant form of modification on mRNA, playing a vital role in many life processes and diseases [26]. However, the function of m6A modification in heat shock-related ESCC progression remains unknown. To explore the role of m6A in this process, we evaluated the heat resistance ability of EC109 and KYSE150 cells with or without m6A writer inhibitor STM2457. We treated cells in a 45 °C hot water bath for 30 min. Through flow cytometry analysis, we confirmed that the inhibition of the m6A modification significantly promote cell death of both EC109 and KYSE150 cells (Fig. 1B-D). The intracellular results show that m6A modification is very important for the response of heat shock.

2.2. Heat shock causes significant alterations in m6A modification

Next, meRIP-seq were performed to study the differences in m6A modifications before and after heat shock. Through high-throughput sequencing, we detected 23,324 m6A modification sites, of which 13,618 sites showed no significant change before and after heat shock. Compared to the control group, 3535 sites were up-regulated and 6171 sites were down-regulated after heat shock (Fig. 2B and table S1). The differences in all genes are shown in the heatmap (Fig. 2C). Further, we analyzed the distribution of m6A modifications on mRNA, the significant differences only found in 3'UTR and CDS (Fig. 2D). After heat shock, the amount of m6A modifications in the 3'UTR significantly increased (2.81 % vs 2.16 %). Peak density analysis showed a significant increase in modification density in the 3'UTR in the heat shock group (Fig. 2E), while the authenticity of these differential m6A modifications was confirmed through the motif GGACU (Fig. 2F). Finally, we performed cluster analysis of genes with m6A modification differences. After heat shock, genes with up-regulated m6A modifications were significantly enriched in pathways in cancer, cAMP signaling pathway, Oxytocin signaling (Fig. 2G), while pathways with significantly down-regulated m6A modifications were mainly concentrated in Endocytosis, Huntington's disease, and Alzheimer's disease pathways (Fig. 2H, table S2). The results suggest that after heat shock, a large of genes show significant differences in m6A modifications, and the genes associated with up-regulated m6A modification sites were related to cancer pathways.

2.3. Heat shock promote hyper-m6A modification

Further, we carried out a joint analysis of the cell's transcriptomic data and m6A-seq data. By sequencing the transcriptome of the same batch of cells treated with heat shock. We found that the number of up-regulated mRNA after heat shock to be 596, down-regulated mRNA 1333, and genes with no difference 24,929 (Fig. 3A). By graphing analysis with the difference in m6A modification sites (Δ m6A), difference in mRNA (Δ mRNA), and significantly up-regulated m6A genes

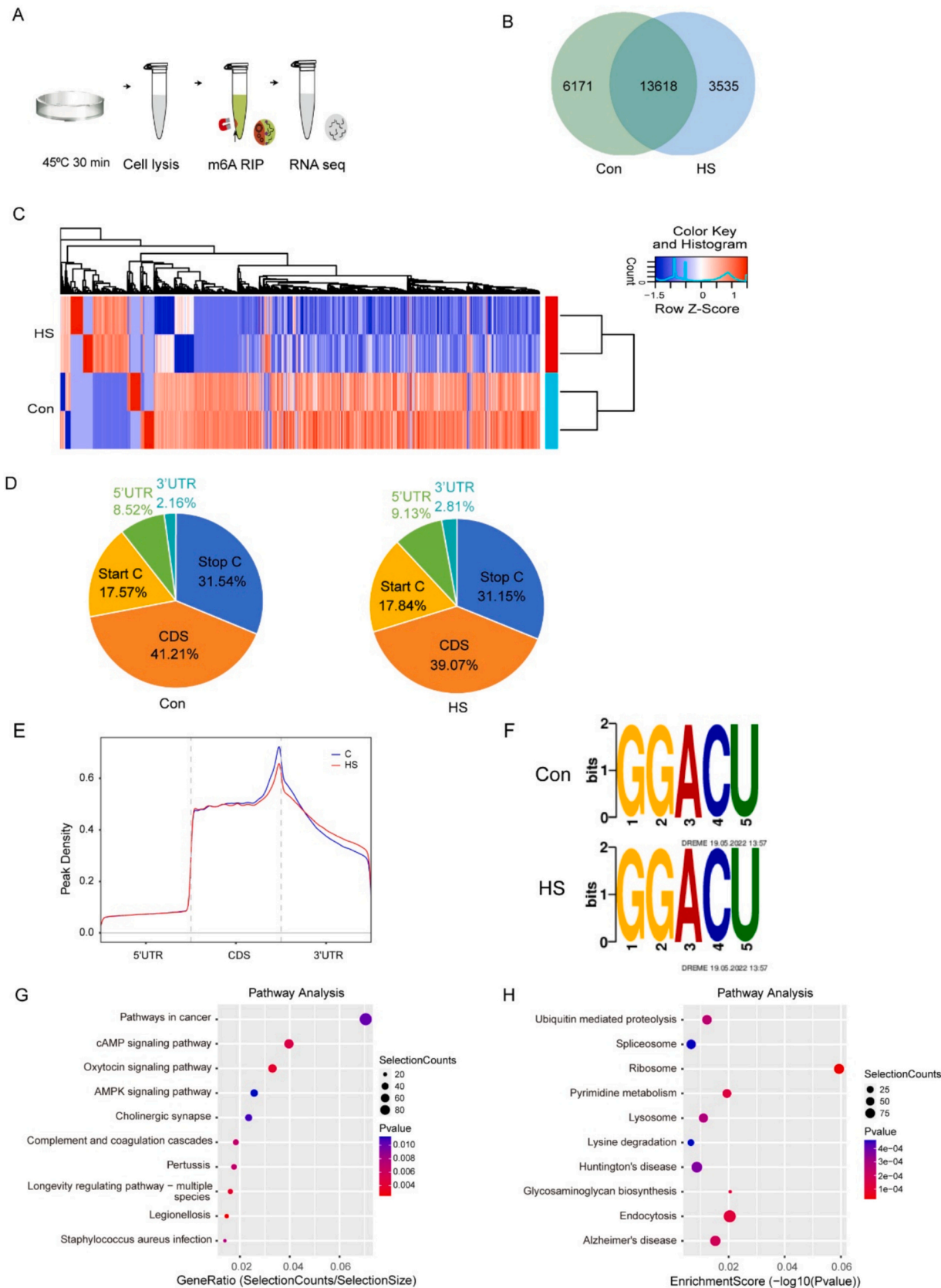


Fig. 2. Differences in m6A modification within cells before and after heat shock for 6 h. (A) Flowchart of heat shock treatment and meRIP-seq ($n = 2$); (B) Ven analysis of m6A modifications before and after heat shock; (C) Heatmap of the significant differences in the m6A modification sites; (D) Pie charts of m6A modification distribution in different mRNA regions before and after heat shock; (E) Peak density analysis of m6A modification distribution in different mRNA regions before and after heat shock; (F) Motif analysis of RNA sequences enriched by m6A antibody before and after heat shock; (G) Cluster analysis of upregulated genes in m6A modification after heat shock; (H) Cluster analysis of genes downregulated in m6A modification after heat shock.

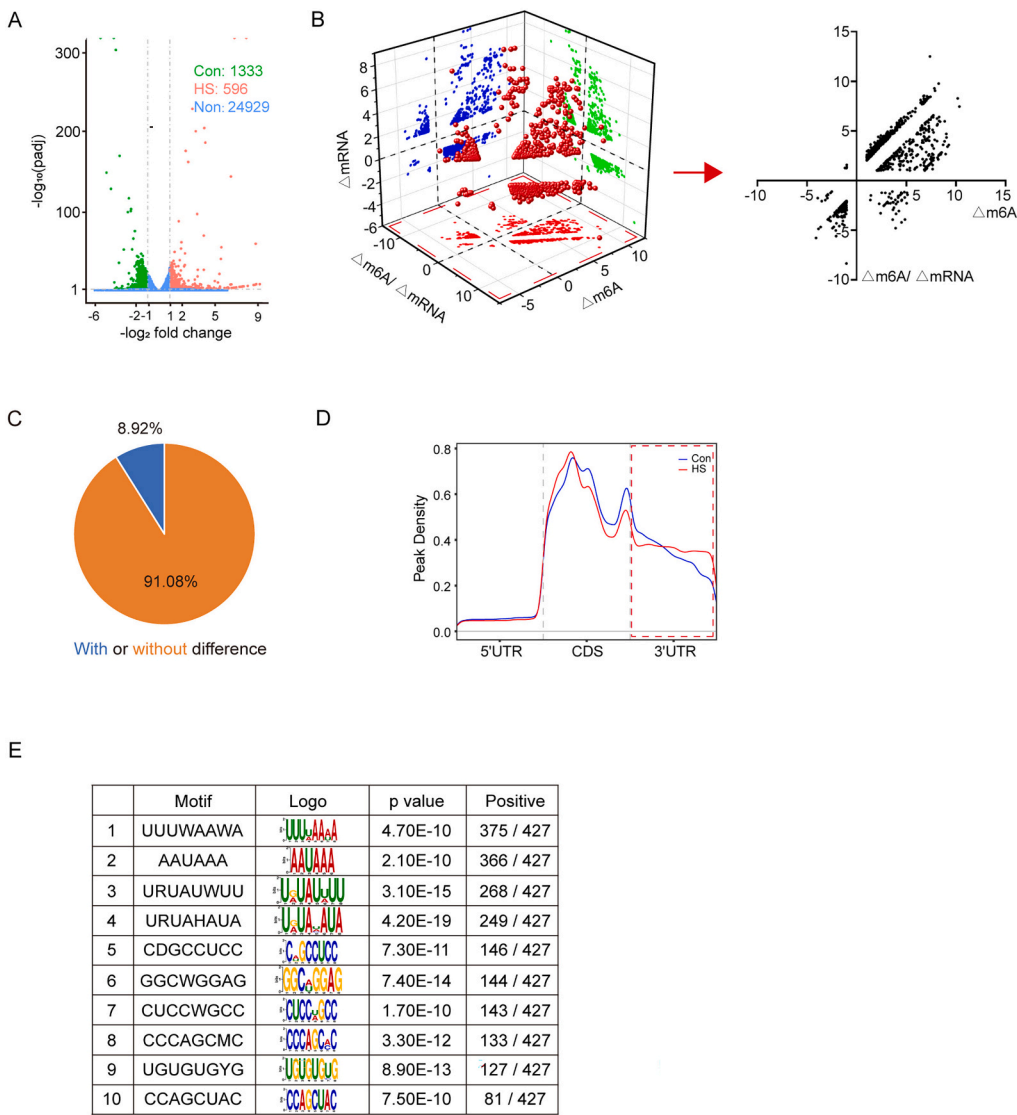


Fig. 3. Joint analysis of meRIP-seq and mRNA-seq data. (A) Volcano plot of differentially expressed mRNA before and after heat shock; (B) Using difference in m6A modification sites (Δm6A), difference in mRNA (ΔmRNA), and significantly up-regulated m6A genes ($\Delta \text{m6A} / \Delta \text{mRNA}$) as the X, Y, Z axis for plotting, to analyze the mRNA where m6A modification has significantly changed before and after heat shock; (C) Percentage of genes with hyper-m6A modification among all detected m6A modifications; (D) Peak density analysis of m6A modification and motif analysis; (E) MOTIF analysis of genes with hyper-m6A modification.

($\Delta \text{m6A} / \Delta \text{mRNA}$) as the X, Y, Z axis, through two-dimensional analysis (X: Δm6A , Y: $\Delta \text{m6A} / \Delta \text{mRNA}$), we found that most of the hyper-m6A modification sites showed a significant up-regulation of Δm6A and $\Delta \text{m6A} / \Delta \text{mRNA}$ (Fig. 3B). We found that 91.08 % of the m6A difference originated from its mRNA difference (after deducting the difference multiple of mRNA, the change in m6A is less than 2 fold). Among all detected m6A modification sites, 8.92 % of m6A modifications show hyper-regulation. (Fig. 3C). The interaction network of core genes was shown in Fig. S1A. The cluster analysis of these genes showed that the most significant pathways to be cellular response to DNA damage stimulus pathway, chromatin binding pathway, and embryonic morphogenesis pathway (Fig. S1B). Peak density analysis showed that compared to over all different m6A modification sites, the difference enrichment of hyper-m6A modification genes in the 3'UTR was more obvious (Fig. 3D). The significant difference in this region suggests that there may be some unknown RNA binding proteins binding to the 3' UTR and regulating the m6A modification. Further, MOTIF analysis showed that the first four of the top five motif were A/U-rich sequences (Fig. 3E). This led us to consider the RNA binding protein HSP70 family, which have such a structure and the ability to bind and stabilize RNA,

but the related mechanism was not yet deeply researched.

2.4. Proteomics identifies m6A 'writer' binding protein after heat shock

To investigate whether there were RNA-binding proteins binding to the m6A 'writer' complex to affect m6A modification after heat shock, we investigated the differences in the binding proteins of METTL3 before and after heat shock through protein co-immunoprecipitation and HPLC-MS/MS methods (Fig. 4A). Surprisingly, after heat shock, the number of proteins that binding to METTL3 were significantly decreased and the types of proteins drastically changed (Fig. 4B). Specifically, before heat shock we detected 86 METTL3 proteins; after heat shock, only 44; (table S3). This suggests that heat shock dramatically changes the binding proteins of the METTL3/14 complex, which may lead to a change in the mode of m6A modification. MCODE analysis showed that HSP70 family proteins (HSPA1A, HSPA1B, HSPA6, HSPA5, HSPD1) were among the binding proteins and occupied relatively central positions among all proteins (Fig. 4C). Cluster analysis also showed that the protein refolding pathway has the most significant clustering (Fig. S2A). Subsequently, we verified the binding of HSP70 and METTL3

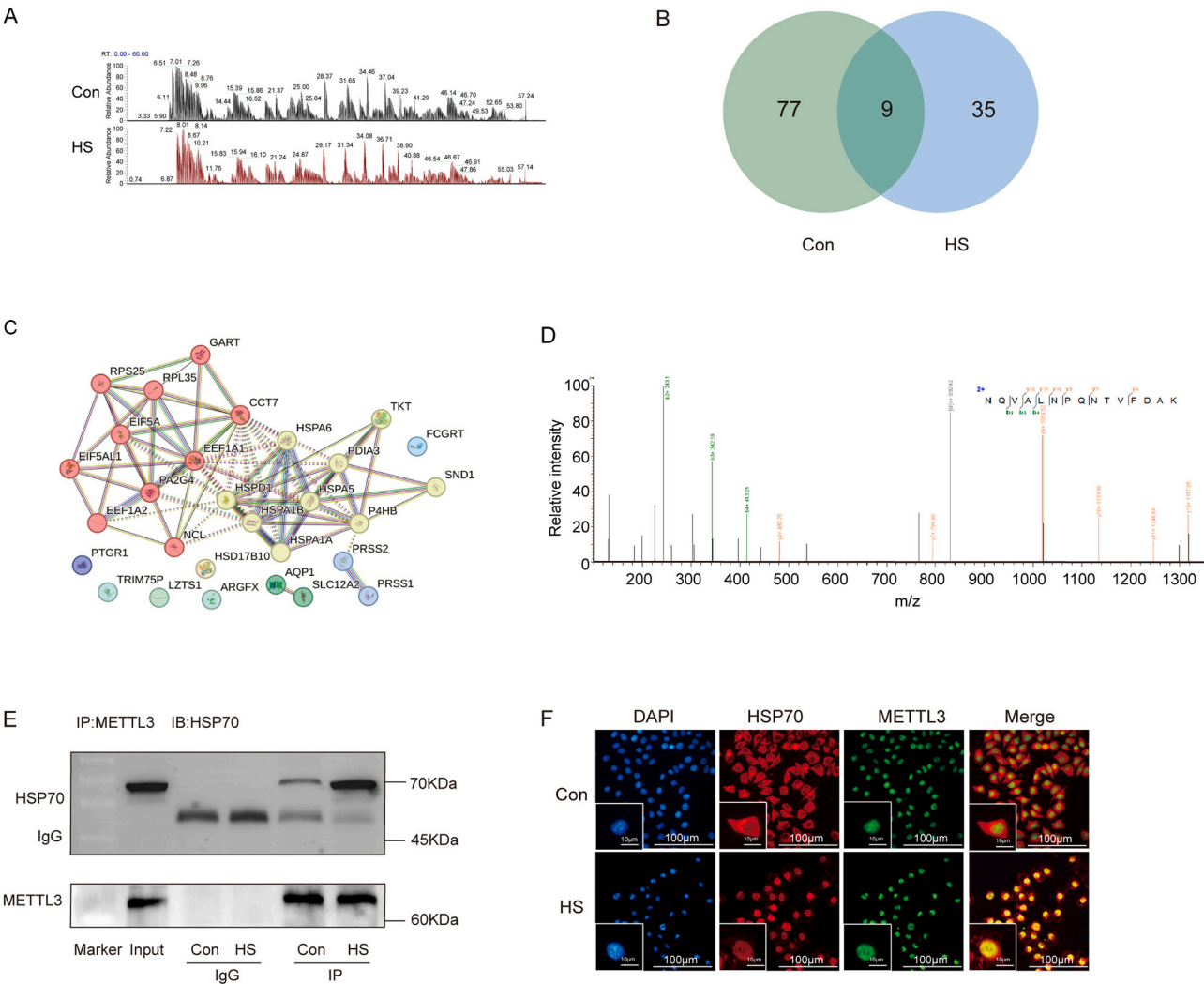


Fig. 4. Proteomics detection of differences in METTL3 binding proteins before and after heat shock. (A) Grouping and peak map of co-IP; (B) Venn analysis of METTL3 binding proteins before and after heat shock; (C) MCODE analysis of METTL3 binding proteins after heat shock; (D) HPLC-MS/MS detection of the HSP70 peak map; (E) Co-IP using METTL3 antibody, western blot to detect the protein level of HSP70 before and after heat shock; (F) Immunofluorescence observation of subcellular localization of HSP70 and METTL3 before and after heat shock, HSP70 was marked with red fluorescence, METTL3 was marked with green fluorescence, and blue is DAPI, marking the cell nucleus. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

proteins. The mass spectrum of HSP70 (HSPA1B) in proteomic detection was shown in Fig. 4D. We confirmed the binding of HSP70 protein and METTL3 after heat shock through co-IP and western blot (Fig. 4E). Immunofluorescence co-location experiments also showed that after heat shock, HSP70 protein entered the nucleus and had the same subcellular location with METTL3 (Fig. 4F).

2.5. Inhibition of HSP70 nuclear translocation reverses the expression and m6A modification of HSPA1B mRNA

The molecular function of the HSP70 protein relies on the conformational change of its ATP structural domain. When we inhibit the entry of HSP70 protein into the nucleus with structural inhibitors (Fig. 5A), the abundance of HSP70 protein bound to METTL3 significantly decreased (Fig. 5B). Next, we used METTL3 and HSP70 antibodies to perform RIP to enrich the mRNA bound to them. Through qPCR, we found that after heat shock, the ability of METTL3 and HSP70 to bind HSPA1B mRNA significantly enhanced, and the use of the HSP70 structural inhibitor significantly inhibited this binding action (Fig. 5C). Based on qPCR of total RNA, the use of inhibitors did not completely inhibit the expression of HSPA1B mRNA (Fig. 5D), while the results of

meRIP-qPCR displayed that the inhibition of HSP70 did not fully abolish the m6A modification of HSPA1B (Fig. 5E). However, the hyper-m6A modification in the mRNA of HSPA1B were abolished. By inhibiting mRNA transcription, we discovered that blocking the nuclear translocation of HSP70 markedly suppressed the mRNA stability of HSPA1B (Fig. 5F).

2.6. RIP-seq reveal the mRNA binding of METTL3 and HSP70 post heat shock

Although preliminary confirmation of genes bound by HSP70 and METTL3 was primarily confirmed through RIP-qPCR, it was insufficient to assert that a variety of genes were jointly bound and modified by HSP70 and METTL3. Subsequently, after 6 h post heat shock, nuclei from heat shocked cells were extracted and subjected to RIP-seq using anti-HSP70, anti-METTL3, and anti-IgG antibodies (Figs. 6A and B) for a comprehensive portrayal of the mRNA jointly bound by HSP70 and METTL3. Findings exhibited 1470 unique mRNA identified by the anti-HSP70 antibody group and 1613 unique mRNA by the anti-METTL3 antibody group post heat shock. Astonishingly, through VEN analysis, the jointly bonded mRNA by METTL3 and HSP70 number at 754,

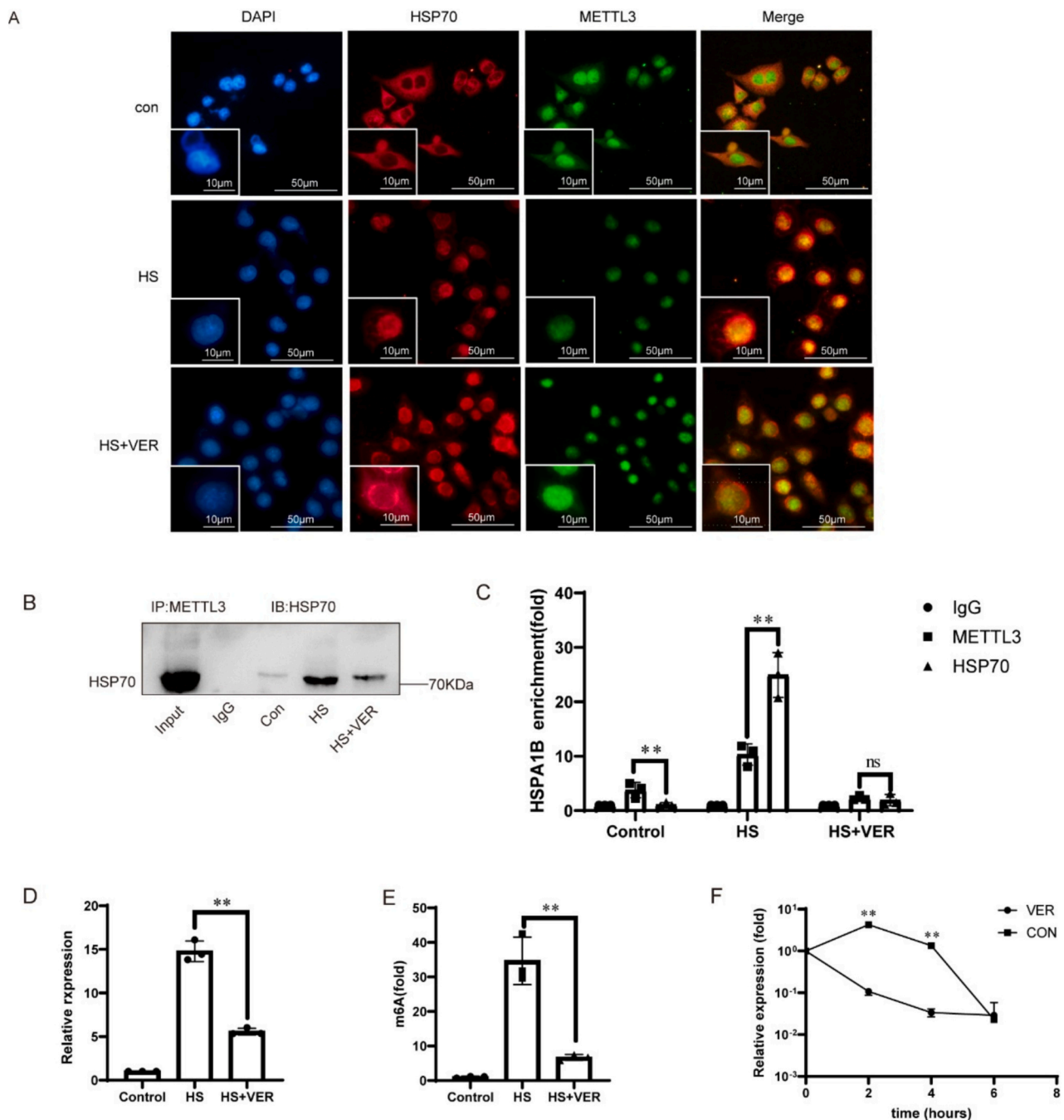


Fig. 5. Detection of m6A modification differences in HSPA1B mRNA after inhibition of HSP70 protein. (A) After adding the HSP70 inhibitor VER, immunofluorescence staining were performed to observe the nuclear entry of HSP70 protein and its subcellular location with METTL3 protein; (B) co-IP detects the binding of METTL3 and HSP70 after adding the HSP70 inhibitor; (C) Using anti-HSP70 antibody and anti-METTL3 antibody for RNA immunoprecipitation follow by qPCR to detect the ability of HSP70 or METTL3 to bind HSPA1B mRNA under different treatment conditions; (D) qPCR detects the effect of the use of the HSP70 inhibitor on the expression of HSPA1B mRNA; (E) Using anti-m6 A antibody for meRIP follow by qPCR to detect the effect of the use of the HSP70 inhibitor on the m6 A modification of HSPA1B mRNA; (F) Following heat shock treatment, VER was administered to inhibit HSP70 nuclear translocation. After a 4-h recovery period in the incubator, transcription was suppressed using actinomycin D. Subsequently, cells were harvested at 2, 4, and 6 h post-transcriptional inhibition, qPCR assays were performed to evaluate HSPA1B mRNA stability. The data were presented as mean $\pm$ SD, one way ANOVA or student *t*-test was used for the analysis, $n = 3$, $^{**}p < 0.01$.

collectively sharing approximately 50 % of detected mRNA (Fig. 6C, table S4). The clustering of these genes revealed the top three pathways to be Cilium, peptidyl-amino acid modification, and protein phosphorylation (Fig. 6D). Venn analysis was performed to identify genes with different m6A modification, hyper-m6A and HSP70/METTL3 binding mRNA (Fig. 6E). Genes with significant changes in m6A modification and METTL3/HAP70 binding were shown at Fig. 6F.

2.7. Stronger nuclear expression of HSP70 predicts poor prognosis for ESCC

Through the heat shock model, we found that HSP70 could participate in mRNA m6A modification in nucleus. We wondered about the subcellular location of HSP70 in tumor tissue and its clinical significance. Immunohistochemical staining results showed that in normal tissues, HSP70 was only observed at the basal layer with all subcellular

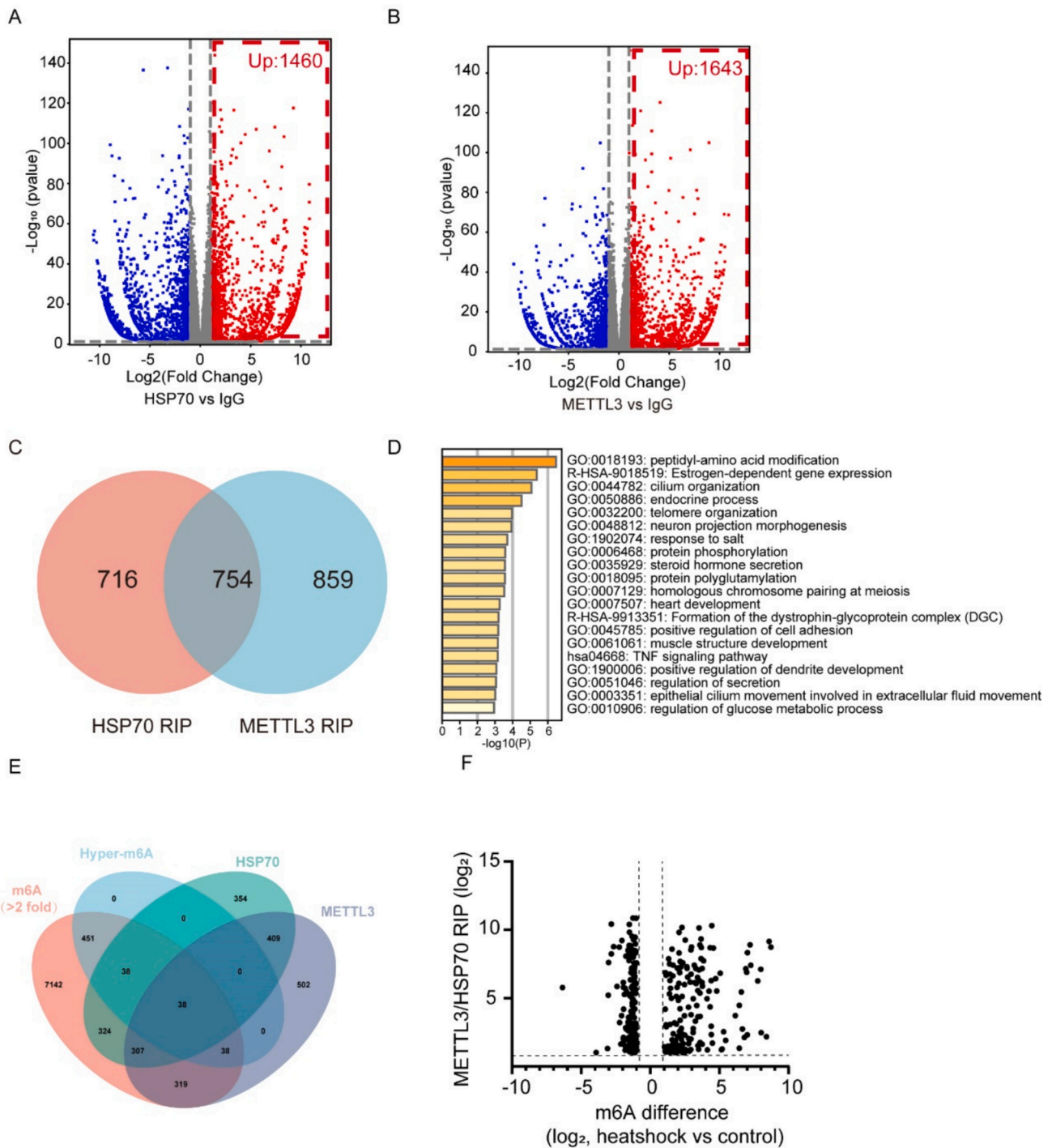


Fig. 6. Identification of mRNA bound by HSP70 and METTL3 via RIP-seq. (A) Volcano plot of mRNA bound by METTL3($n = 2$); (B) Volcano plot of mRNA bound by HSP70($n = 2$); (C) Venn analysis of mRNA bound by both HSP70 and METTL3; (D) Heatmap of jointly bonded mRNA; (E) Cluster analysis of mRNA jointly bound by HSP70, METTL3 and gene with up regulated m6A or hyper-m6A modification; (F) Volcano plot of gene significant changes in m6A modification and METTL3/HAP70 binding.

locations being the nucleus. In some dysplasia tissues, HSP70 showed positive staining in the epithelial layer, but the positive signals in the epithelial layer were found in the cytoplasm. In tumor tissues, HSP70 showed various staining results, but nuclear localization was present in most samples, and in some samples, different levels of expression could also be detected in the cytoplasm (Fig. 7A). Overall, compared to normal tissues, tumor tissues had stronger stain in nucleus (Fig. 7B). The scores of the paired cancer and normal tissues were shown in Fig. 7C. Finally, the prognosis analysis indicated that patients with high expression of

HSP70 in the nucleus had a poorer prognosis (Fig. 7D). Detailed clinical data related to the tissue microarray can be found in Table S5.

3. Discussion

In general, only 16 % of genes in the human genome contain A/U rich elements in the 3' UTR region of their mRNA [27]. However, we observed marked enrichment of these motifs in hyper-m6A-modified mRNAs post-heat shock (Fig. 3E), prompting investigation into HSP70

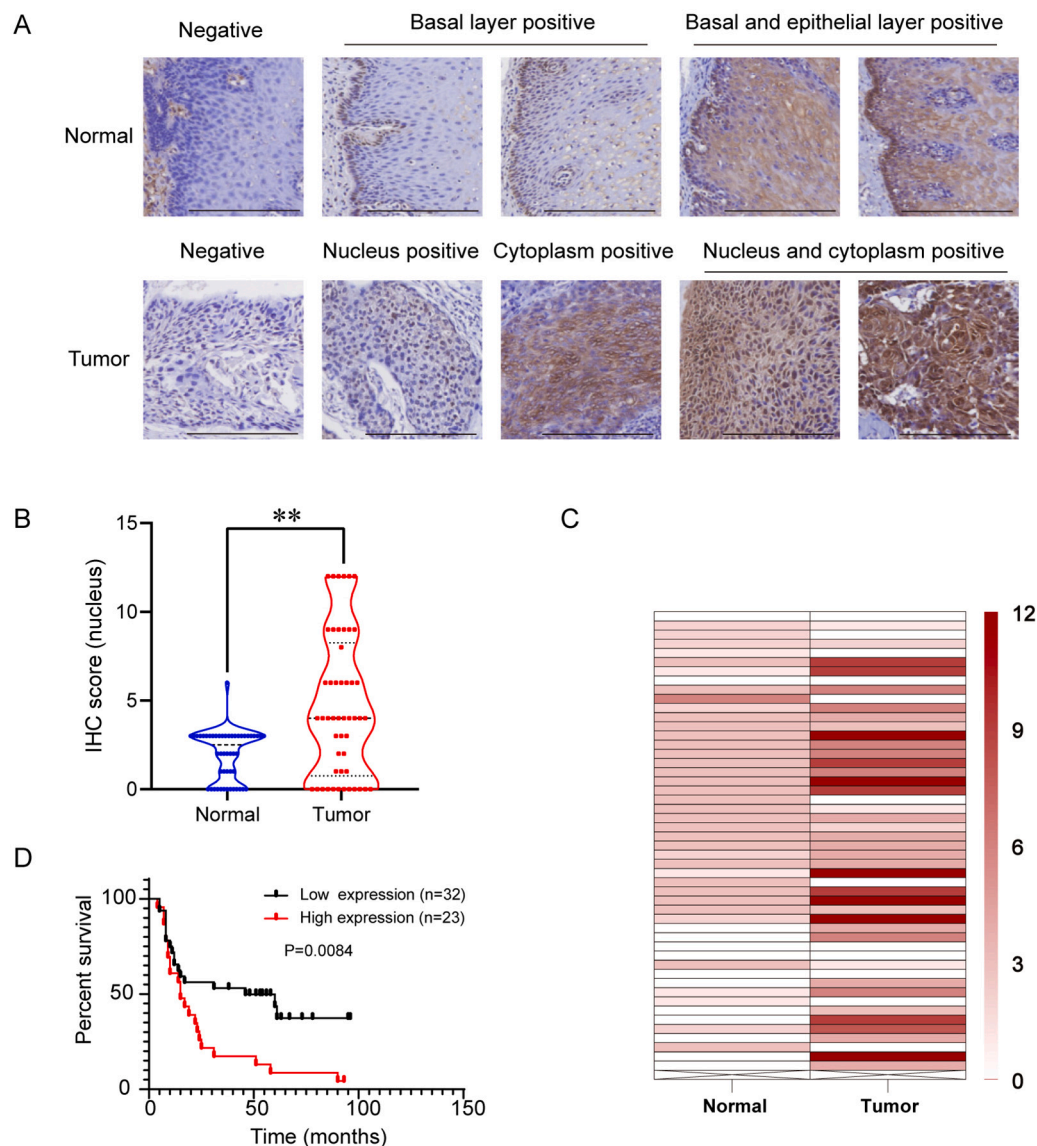


Fig. 7. Staining of HSP70 protein in ESCC tissues and its clinical significance. (A) Immunohistochemistry staining of HSP70 protein in adjacent tissues and cancerous tissues ($n = 55$), scale bar 200 μm ; (B) Scoring of the staining intensity of HSP70 in the cell nucleus in adjacent tissues and cancerous tissues; (C) Paired display of scoring of the staining intensity of HSP70 in the cell nucleus in adjacent tissues and cancerous tissues; (D) Analysis of the correlation between the staining intensity of HSP70 in nucleus and the prognosis of ESCC patients. Students t -test was used for the analysis, $**p < 0.01$.

– a known A/U-rich sequence binder [28], as a candidate guide. Crucially, HSP70's tumor-associated nuclear localization (Figs. 4F, 7A), contrasting with cytoplasmic predominance in cultured cells, creates spatial prerequisites for METTL3/14 complex recruitment via its peptide-binding domain, a mechanism independent of its ATP-dependent chaperone activity [29,30]. This compartmentalized operation explains how HSP70 stabilizes oncogenic transcripts [31–33] in malignancies while maintaining cytoprotective roles in normal stress responses, bridging its dual association with poor cancer prognosis [34] and cellular recovery. Importantly, our RIP verification of nuclear HSP70-METTL3/14-mRNA tripartite complexes (Fig. 5) mechanistically links prior observations of HSP70's RNA-binding capacity [28,35] to epitranscriptomic regulation, demonstrating that stress-triggered nuclear translocation provided a new insight of HSP70 as a regulator of m6A pathway. The requirement for nuclear localization, overlooked in earlier cytoplasmic-focused studies [31–33], recontextualizes HSP70's therapeutic targeting potential.

Every ~57 nucleotides in mRNA, there is a consensus sequence DRACH which means that each transcript will have numerous potential

methylation sites. However, only a small fraction of these DRACH sequences were actually methylated. The mechanisms behind the specific methylation is not well comprehended [36]. Previous studies have identified several adaptors for METTL3 and METTL14, including WTAP, VIRMA, RBM15/15B, ZC3H13, among others [37]. Their depletion could result in an approximately 80 % loss of m6A modifications [38], demonstrating that these widespread binding proteins of METTL3/14 complex do not possess the capacity to selectively modify specific mRNAs with m6A. Some research indicate that transcription factors or proteins associated with RNA polymerase could recruit the METTL3/14 complex to the transcription complex, such as smad2/3, CEBPZ, collectively transcribing and modifying no more than 200 genes [17] [39]. Some epigenetic studies have also shown that METTL3/14 complex could directly binds to H3K36me3 or H3K27, which guides the deposition of substantial m6A modifications [18,40]. However, at present, our understanding of m6A methylation preference still limited. It is likely that understanding the different m6A writer components and their regulation will be key to understanding the nature and function of regulated m6A sites [36].

In addition to esophageal heat shock and heat stroke, another common scenario of heat shock in clinical practice is radiofrequency ablation used for solid tumor therapy. Incomplete radiofrequency ablation could paradoxically enhance tumor angiogenesis [41], intratumoral metastasis [42] and distal metastasis [25], which significantly limiting the application of this technique. The concurrent use of inhibitors targeting PI3K/Akt or HSP70 during radiofrequency ablation significantly impeded tumor proliferation and distal metastasis [41], albeit the precise mechanism remains unclear. Our research provides valuable insight into the vital role of HSP70 in this context. In esophageal-related studies, radiofrequency ablation is utilized to ablate the dysplasia of Barrett's esophagus [43] and clearing esophageal squamous epithelial cell neoplasia [44]. Clinical studies targeting ESCC also demonstrate high safety and good efficacy [45]. The use of HSP70 inhibitors prior to radiofrequency ablation may bring additional benefits to patients.

In conclusion, our study found that after heat shock, HSP70 protein could binds to the m6A 'writer' complex and guides the m6A modification of heat shock-related genes. It will enrich our understanding of the m6A's "writing" mode and provides an additional understanding of the molecular mechanism by which HSP70 protein binds to and stabilizes mRNA. It provides a new perspective for the study of molecular mechanisms related to cancer and heat shock relative diseases.

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CRedit authorship contribution statement

Bin Du: Investigation. **Jia Wang:** Writing – original draft, Visualization. **Jun Ma:** Writing – review & editing, Funding acquisition, Data curation. **Pu Wang:** Writing – original draft, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no conflicts of interest with the contents of this article.

Data availability

All datasets are included within the article or are available from the corresponding author: Pu Wang (372746529@qq.com).

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